

Flight Booking Prediction Model

Leveraging machine learning to identify high-intent customers and transform airline marketing strategy





Executive Summary

Model Performance

RandomForest Classifier with 100 trees achieves 77.74% recall rate, successfully identifying nearly 8 out of 10 customers who will complete bookings.

Key Predictor

Booking origin popularity emerged as the strongest signal at 19.25% feature importance, revealing geographic markets as critical conversion drivers.

Business Impact

Model enables proactive marketing campaigns by identifying 78% of potential bookings before customers reach the airport, transforming reactive sales strategies.

77.74% 76.61% 19.25%

Recall Rate

Successfully captures potential bookings

ROC-AUC Score

Strong discriminative ability

Top Feature

Origin popularity importance

Project Overview

Business Challenge

The airline industry has fundamentally shifted. Today's customers research extensively and book flights online before ever stepping into an airport. Traditional reactive sales strategies—waiting for customers to arrive at booking counters—no longer capture the full market opportunity. Airlines need predictive intelligence to engage customers early in their decision journey, delivering personalized offers at the optimal moment when booking intent is highest.



Data Exploration Findings

Dataset Quality

Total Records

50,000 customer interactions analyzed

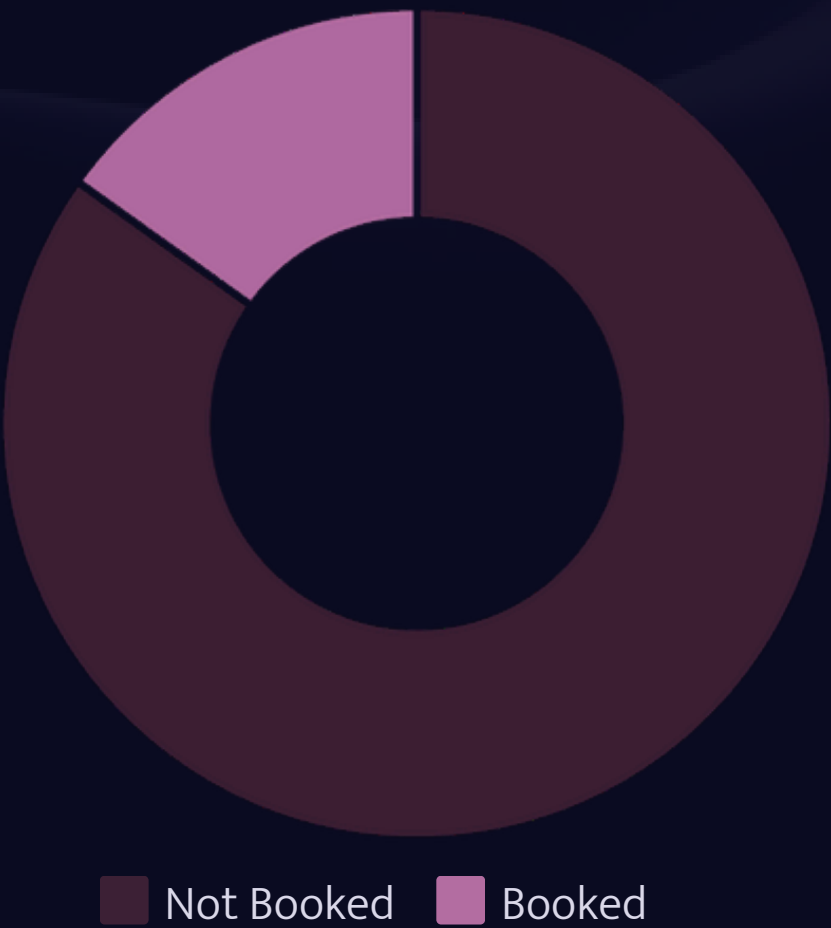
Data Completeness

Zero missing values—clean, reliable dataset

Booking Rate

14.96% conversion rate (7,478 bookings)

Class Imbalance Challenge



The dataset exhibits a **5.69:1** imbalance ratio, with non-bookings significantly outnumbering completed bookings. This mirrors real-world airline conversion rates but requires strategic model optimization.

📌 **Strategic Decision:** Applied `class_weight='balanced'` parameter to ensure the model learns effectively from the minority class (bookings), preventing it from simply predicting "no booking" for all cases.

Feature Engineering Strategy

"Raw data rarely tells the complete story. Strategic feature engineering transforms hidden patterns into predictive power."

We created 15 sophisticated features designed to capture customer behavior patterns, travel preferences, and booking urgency signals. This engineering process expanded our predictive capability from 14 original attributes to 934 encoded features.

Temporal Patterns

is_weekend_flight: Identifies Saturday/Sunday departures to capture leisure vs. business travel behavior patterns.

Booking Urgency

booking_lead_category: Segments customers from last-minute bookers to very long advance planners, revealing purchase intent.

Trip Duration

stay_category: Classifies short getaways to extended trips, indicating travel purpose and customer segment.

Customer Value

total_extras: Counts additional services (0-3) like meals, baggage, and seat upgrades as value indicators.

Route Intelligence

route_popularity: Measures frequency of specific routes to identify high-demand destinations and seasonal patterns.

Travel Groups

is_family_trip: Flags groups of 3+ passengers to distinguish family travel from business or solo journeys.

Model Selection: RandomForest Classifier



Why RandomForest?

We selected RandomForest Classifier as our predictive engine because it uniquely addresses the complex requirements of airline booking prediction while maintaining business interpretability.

Model Configuration

Business Interpretability

Provides clear feature importance scores, enabling stakeholders to understand why customers book and make data-driven decisions.

Imbalance Handling

Built-in class weighting mechanism ensures the model learns effectively from our 85/15 booking imbalance ratio.

Data Flexibility

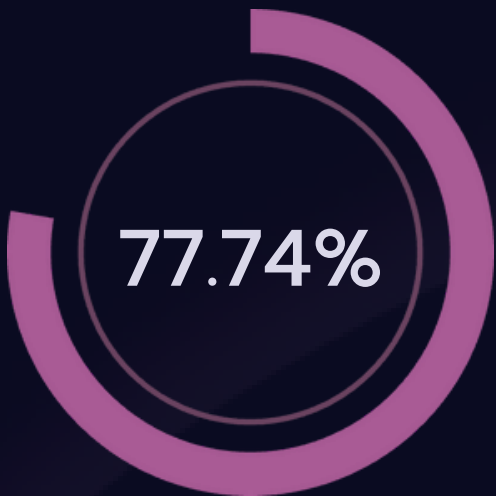
Seamlessly processes mixed categorical and numerical data without requiring extensive preprocessing or scaling.

Robust Performance

Resistant to outliers and noise in real-world data, delivering consistent predictions across varying conditions.

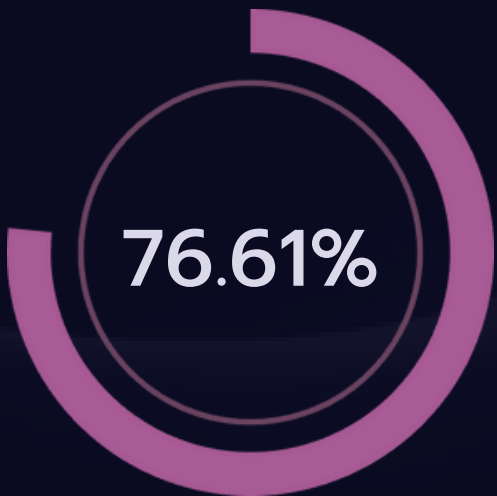
Model Performance Results

Evaluated on 10,000 held-out test samples, our model demonstrates strong predictive capability with strategic emphasis on capturing booking opportunities.



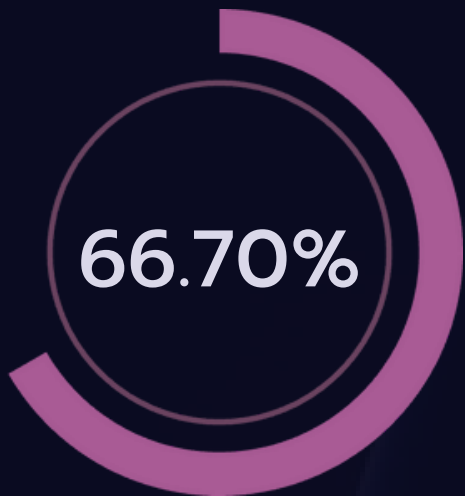
Recall

Successfully identifies bookings



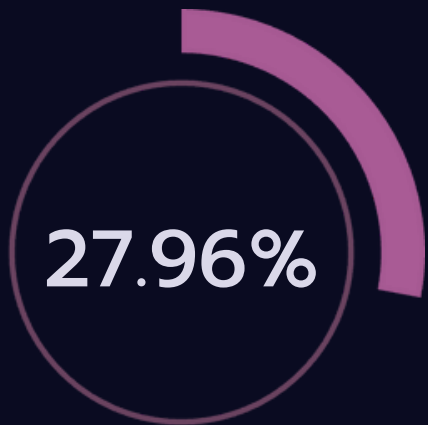
ROC-AUC

Discriminative ability



Accuracy

Overall correct predictions



Precision

Booking prediction accuracy

Why Recall Matters More Than Accuracy

78 out of 100

Our model catches 78% of actual bookings, this is what drives revenue.

The Accuracy Trap

A model with 85% accuracy could simply predict "no one will book" and be correct 85% of the time (matching our class distribution). This high accuracy is useless for business—it catches only 4 bookings out of 100.

85%

Accuracy

Says "no" to everyone

3.8%

Recall

Misses 96 bookings

0

Value

No actionable insights

Our Strategic Choice

We optimized for **recall** because missing potential bookings means lost revenue, while false positives represent only minor marketing costs. Better to contact 100 people and secure 28 bookings than miss 78 opportunities.

67%

Accuracy

Balanced predictions

77.7%

Recall

Captures opportunities

High

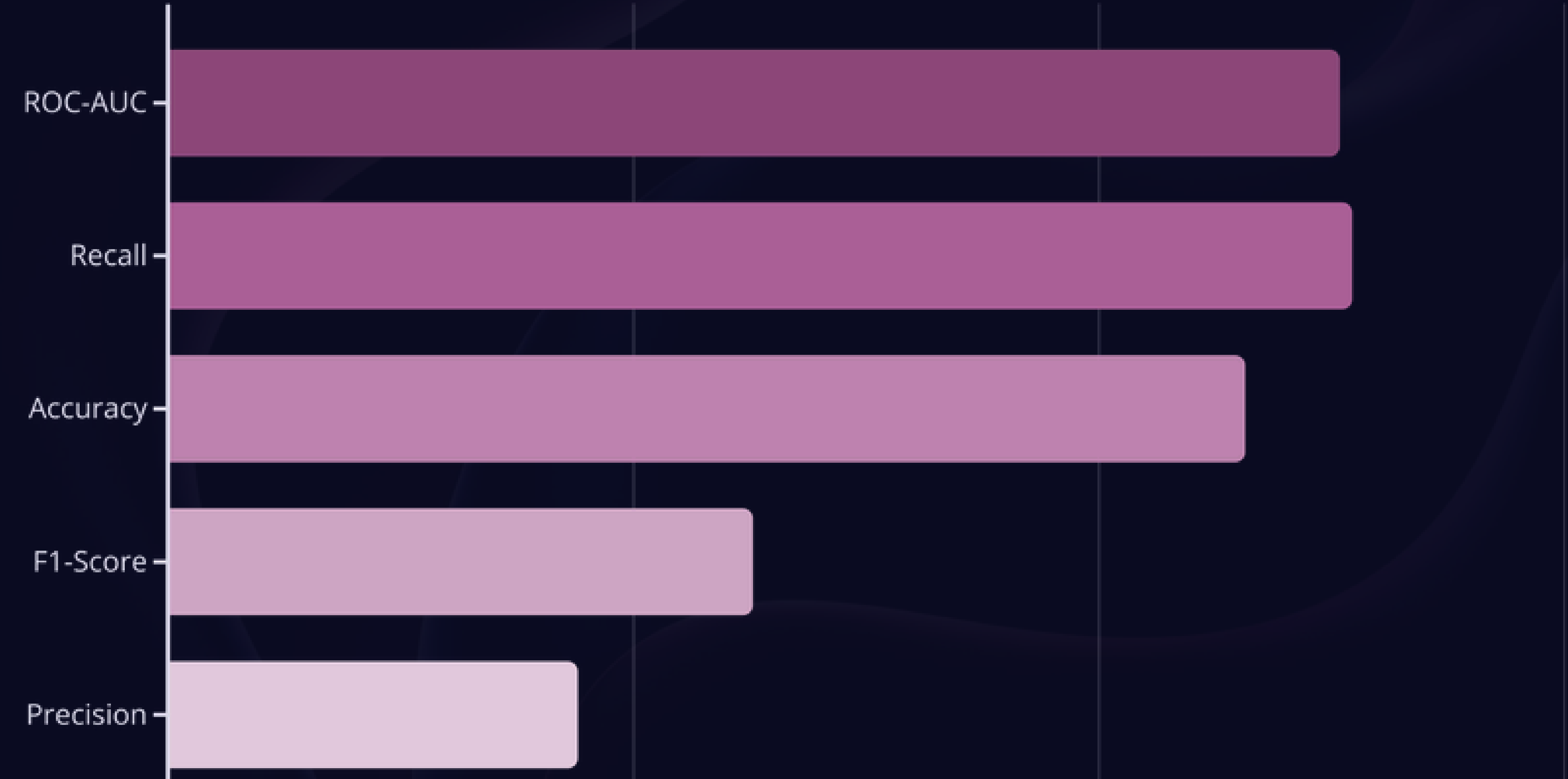
ROI

Actionable targeting

📌 **Business Translation:** Missing a booking opportunity costs the airline a ticket sale. Contacting a customer who doesn't book costs only marginal marketing spend. The economics clearly favor high recall.

Cross-Validation Analysis

We conducted 5-fold cross-validation to assess model stability across different data subsets, testing performance consistency and generalization capability.



Top Predictive Features

Feature importance analysis reveals which customer attributes most strongly predict booking completion, providing clear direction for marketing strategy.



Origin Popularity – 19.25%

The single strongest predictor. Booking location popularity accounts for nearly one-fifth of the model's predictive power, indicating certain geographic markets convert at significantly higher rates.



Australian Customers – 12.62%

Customers booking from Australia demonstrate distinct conversion patterns, making this market a high-priority target for campaigns.



Malaysian Customers – 9.99%

The Malaysian market emerges as another key conversion driver, representing nearly 10% of predictive importance.



Flight Duration – 4.79%

Length of flight influences booking decisions, likely correlating with trip purpose and customer commitment level.



Route Popularity – 4.46%

Frequently traveled routes show higher conversion rates, suggesting demand concentration on established corridors.

Strategic Takeaway

Geographic origin dominates booking prediction—certain markets convert at rates far exceeding others. Marketing resources should prioritize these high-conversion geographies.